Quantum Mechanics Homework 8

Lectures 17–18

Instructions. Show your work and briefly explain your answers.

Useful formulas. You may use the following without derivation. For $\phi = Pe^{ikx} + Qe^{-ikx}$ and a single interface from region i to region j, with real positive wave numbers,

$$j = \frac{\hbar k}{m}\big(|P|^2 - |Q|^2\big), \qquad \cosh^2 u - \sinh^2 u = 1,$$

$$r_{ij} = \frac{k_i - k_j}{k_i + k_j}, \qquad t_{ij} = \frac{2k_i}{k_i + k_j}, \qquad \sum_{n=0}^{\infty} z^n = \frac{1}{1-z} \quad (|z| < 1).$$

Problem 1. Matching at a finite barrier [20 points]

Consider a particle incident from the left on

$$V(x) = \begin{cases} 0, & x < 0, \\ V_0, & 0 < x < L, \\ 0, & x > L, \end{cases} \qquad V_0 > 0, \quad L > 0, \quad 0 < E < V_0.$$

Define $k = \sqrt{2mE}/\hbar$ and $\kappa = \sqrt{2m(V_0 - E)}/\hbar$. Set the incident amplitude to one.

(a) Write the stationary solution in each region using r for the reflected amplitude, t for the transmitted amplitude, and C, D for the two interior coefficients. Explain why *both* interior exponentials must be retained. (4 points)

(b) Write all four matching conditions at $x = 0$ and $x = L$. (5 points)

(c) Now take $E = V_0/2$, so that $\kappa = k$. Solve the matching equations for r and t, retaining the complex phase of t. *Hint: first express Ce^{kL} and De^{-kL} in terms of te^{ikL}.* (7 points)

(d) For this energy, calculate R and T from the exterior currents and verify $R + T = 1$. Find the leading form of T when $kL \gg 1$, using $\cosh(kL) \simeq e^{kL}/2$. (4 points)

Problem 2. Multiple reflections and well resonances [20 points]

Consider the attractive square well

$$V(x) = \begin{cases} 0, & x < 0, \\ -V_0, & 0 < x < L, \\ 0, & x > L, \end{cases} \qquad V_0 > 0.$$

For left incidence at $E > 0$, define

$$k = \frac{\sqrt{2mE}}{\hbar}, \qquad k_2 = \frac{\sqrt{2m(E + V_0)}}{\hbar}.$$

Label the left exterior, interior, and right exterior as Regions 1, 2, and 3. The incident amplitude at $x = 0$ is one. Define $\widetilde{t}$ as the total transmitted amplitude evaluated at $x = L$, so the right exterior wave is $\widetilde{t}\, e^{ik(x-L)}$.

(a) State the energy range for bound states and the energy range for scattering states, relative to the exterior potential. For $E > 0$, explain why the wavelength inside the well is shorter than outside. (3 points)

(b) Calculate t_{12}, t_{23}, and the two reflection amplitudes $r_{21} = r_{23} \equiv \rho$ for waves incident from inside the well. Write the transmitted path amplitudes $\mathcal{A}_0$ for a direct crossing and $\mathcal{A}_1$ for one additional internal round trip. (5 points)

(c) Sum the amplitudes for all possible numbers of internal round trips to find $\widetilde{t}$. Using $t_{12}t_{23} = 1 - \rho^2$, derive

$$T = |\widetilde{t}|^2 = \left[1 + \frac{4\rho^2}{(1-\rho^2)^2} \sin^2(k_2 L)\right]^{-1}.$$

Explain why adding the individual path probabilities instead would miss the resonance. (7 points)

(d) Find the condition for $T = 1$ and the corresponding positive resonance energies E_n. Define $E_L = \pi^2\hbar^2/(2mL^2)$. For $V_0 = 2E_L$, find the smallest allowed n and the lowest positive resonance energy. Is this resonant state a bound state? Explain. (5 points)

Problem 3. The S-matrix and coherent incidence [20 points]

A localized scatterer has the same exterior potential on both sides. At a fixed energy, write

$$\phi(x) \sim \begin{cases} Ae^{ikx} + Be^{-ikx}, & x \to -\infty, \\ Ce^{ikx} + De^{-ikx}, & x \to +\infty, \end{cases} \qquad k > 0.$$

Two separate unit-amplitude incidence experiments give

$$r_L = r_R = \frac{i}{\sqrt{2}}, \qquad t_L = t_R = \frac{1}{\sqrt{2}}.$$

These amplitudes use the same phase convention as the exterior waves above.

(a) Identify the incoming and outgoing coefficients by their propagation directions. Construct the matrix S in the convention

$$\begin{pmatrix} B \\ C \end{pmatrix} = S \begin{pmatrix} A \\ D \end{pmatrix}.$$

Explain what each column represents. (6 points)

(b) Find the reflection and transmission probabilities for incidence from the left alone and from the right alone. (4 points)

(c) Now let $A = 1$ and $D = e^{i\theta}$, with a controlled relative phase θ. Find $B(\theta)$ and $C(\theta)$. Calculate the fractions of the *total incoming current* leaving on the left and right,

$$f_{\mathrm{L}} = \frac{|B|^2}{|A|^2 + |D|^2}, \qquad f_{\mathrm{R}} = \frac{|C|^2}{|A|^2 + |D|^2},$$

and verify $f_{\mathrm{L}} + f_{\mathrm{R}} = 1$. (10 points)